

SHREYASH SANJAY AGRAWAL

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PROFESSIONAL SUMMARY

Data science graduate student (M.S. Business Analytics & AI, UT Dallas, May 2026) who builds machine learning models that answer practical business questions — which customers are likely to default, what an image contains, what text comes next — and explains the results in plain language. Experienced across the full project lifecycle: framing the problem, preparing data, building and validating models, and presenting findings to non-technical decision-makers.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Scikit-learn, XGBoost, SHAP, TensorFlow, Keras), SQL

Machine Learning: Gradient Boosting, Regression, Neural Networks (CNN, LSTM, GRU), Feature Engineering, Hyperparameter Tuning, Model Evaluation (ROC-AUC, A/B Testing, Hypothesis Testing)

Explainability & Communication: SHAP model explanations, data storytelling, translating model results into business recommendations

Tools: Jupyter, Google Colab, Git, GitHub

PROJECT EXPERIENCE

Credit Risk Prediction — Helping Lenders Decide Who to Approve

Jan 2025 – Apr 2025

Python, XGBoost, Scikit-learn, SHAP

- Built a model that predicts which loan customers are likely to default, using 1M+ customer records and 54 engineered behavioral signals; reached 94% accuracy on customers the model had never seen.
- Made the model explainable for business users: SHAP visualizations show exactly why each account was flagged, which is critical for lending decisions that must be justified to customers and regulators.
- Validated reliability with a leakage-proof customer-level data split and two independent holdout sets, so reported performance reflects how the model would behave in production.
- Simulated lending strategies from the model's scores: the strongest option projected a 73% revenue lift while keeping default rates under 7%; packaged the final model for deployment.

Image Classification with Deep Learning — Teaching Computers to See

Sep 2025 – Nov 2025

Python, TensorFlow, Keras, Keras Tuner

- Trained a convolutional neural network to automatically recognize objects across 60,000 images. The neural network is the same core technique behind automated product tagging, visual quality inspection, and content moderation.
- Diagnosed where the baseline model failed (e.g., confusing cats with dogs) and corrected overfitting using a deeper architecture, dropout, weight regularization, and image augmentation.
- Ran a systematic, automated hyperparameter search instead of trial-and-error, lifting test accuracy from 70.4% to 80.1%, which is a ~10-point improvement from disciplined experimentation.

Text Generation with Neural Networks — The Technique Behind Autocomplete

Sep 2025 – Nov 2025

Python, TensorFlow, Keras, NLTK

- Built a character-level language model on the full text of *Pride and Prejudice*. The language model is the foundational approach behind autocomplete, predictive text, and modern generative AI.
- Compared four neural architectures (single- and 2-layer LSTM, GRU, regularized LSTM) to quantify trade-offs between accuracy, training cost, and overfitting; best model reached 2.99 validation perplexity.
- Measured output quality with multiple complementary metrics (perplexity, BLEU, ROUGE, diversity scores) and built a temperature-controlled sampler to balance creativity against coherence.

EDUCATION

The University of Texas at Dallas — M.S., Business Analytics and Artificial Intelligence

May 2026

Narsee Monjee University, Mumbai — B.Tech., Computer Science and Business Systems

May 2023